Machinery Safety Specialist (CE)

Tab 3: Risk assessment
Module 2: Risk estimation (methods)

Tab3-2/2 20220901

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Step 3: Estimating risks

EN-ISO 12100

- You can decide for yourself which risk assessment methodology is appropriate for the application.

- In this module we cover:
  - Risk graph (various variants), origin prEN1050
  - Kinney & Wiruth (sometimes called Fine & Kinney)
  - Risk matrix
  - Numerical scoring
  - Hybrid tool

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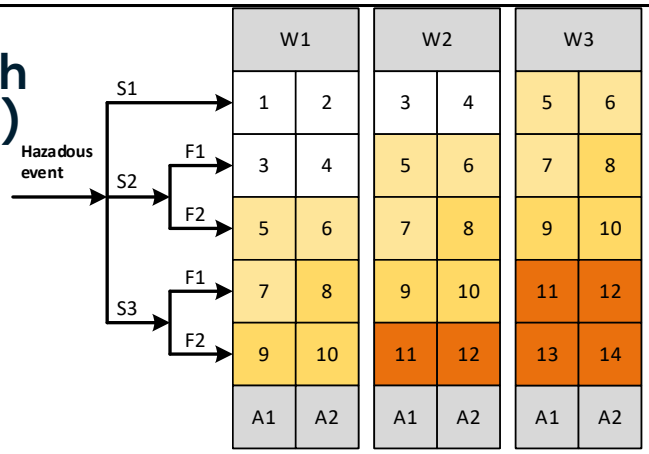
Step 3: Estimating risks

EN-ISO 12100

- For risk assessment based on the Machinery Directive, a risk graph methodology is often used.
- Kinney & Wiruth is often used for risk assessment based on RI&E (e.g., risk assessment based on the Work Equipment Directive)

Risk graph (14-point)



Risk	Risk-class
1-4	*Low risk*
5-7	*Possible risk*
8-10	*High risk*
11-14	*Very high risk*

S Severity of injury
S1 Minor injuries (recoverable)
S2 Mayor injuries (irreparably)
S3 One ore more deaths

P Probability of event
P1 Low
P2 Avarage
P3 High

F Frequency and exposure
F1 Seldom to regularly (2x/ 8 hours) and / or short exposure (15 min. / 8 hours)
F2 Often to continuous (> 2x/ 8 hours) and /or longer exposure (> 15 min. / 8 hours)

A Avoidance
A1 possible under certain circumstances
A2 Hardly possible

Tab3-2/2-Sh.5

Exercise 5

- Carry out a risk assessment for (one of) the pedestal drilling machines using the risk graph method.
- Use the entry sheets at the back of Tab 3 of the map.
- Plenary discussion.

Tab3-2/2-Sh.6

Exercise 5

Kinney and Wiruth (Linear method)

Severity of injury (S)	x	Frequency & Exposure (F)	x	Probabi-lity (P)	=	Risk (R)

R	Risk
<20	*Low risk*
20-69	*Possible risk*
70 - 199	*Considerable risk*
200 - 399	*High risk*
>400	*Very high risk*

S Severity of injury
1 Minor (no absenteeism)
3 Considerable (with absenteeism)
7 Severe
15 Deathly (one death)
40 Disaster (several death)
100 Catastroph (more death)

F Frequency
0,5 Very rare (yearly)
1 Rare (Few times a year)
2 Monthly
3 Weakly
6 Daily
10 Continously

P Probability
0,1 Almost unimaginable
0,2 Almost impossible
0,5 Very unlikely
1 Borderline between unlikely and possible
3 Quite possible
10 To be expected

Exercise 6

- Copy the hazards of exercise 4 onto the entry sheet.
- Carry out the risk assessment for these hazards using the Kinney & Wiruth method.

- Plenary discussion.

Exercise 6

Step 3: Estimating risks

Risk matrix (NPR-ISO/TR 14121-2, paragraph 6.2)

Table 1 — Risk estimation matrix (example)

Probability of occurrence of harm	**Severity of harm**			
	Catastrophic	Serious	Moderate	Minor
Very likely	**High**	**High**	**High**	**Medium**
Likely	**High**	**High**	**Medium**	**Low**
Unlikely	**Medium**	**Medium**	**Low**	Negligible
Remote	**Low**	**Low**	Negligible	Negligible

Tab3-2/2-Sh.11

Step 3: Estimating risks

Numerical scoring (NPR-ISO/TR 14121-2, paragraph 6.4)

The slides show an example of the numerical scoring method.
Other interpretations are also possible, however not covered in the standard.

Example:

- Severity score (SS)
- Probability score (PS)

- Risk score (RS) = SS + PS

Tab3-2/2-Sh.12

Step 3: Estimating risks

Numerical scoring (NPR-ISO/TR 14121-2, paragraph 6.4)

Severity score (SS)

- catastrophic SS = 100,
- serious 99 ≥ SS ≥ 90,
- moderate 89 ≥ SS ≥ 30,
- minor 29 ≥ SS ≥ 0.

Step 3: Estimating risks

Numerical scoring (NPR-ISO/TR 14121-2, paragraph 6.4)

Probability score (PS)

- very likely PS = 100 likely or certain to occur,
- likely 99 ≥ PS ≥ 70 can occur (but not probable),
- unlikely 69 ≥ PS ≥ 30 not likely to occur,
- remote 29 ≥ PS ≥ 0 occurrence so remote as to be essentially zero.

Step 3: Estimating risks

Numerical scoring (NPR-ISO/TR 14121-2, paragraph 6.4)

Table 2 — Risk score categories used

—	**high**	≥ 160
159 ≥	**medium**	≥ 120
119 ≥	**low**	≥ 90
89 ≥	**negligible**	≥ 0

Tab3-2/2-Sh.15

Step 3: Estimating risks

Hybrid method (NPR-ISO/TR 14121-2, paragraph 6.5)

The hybrid method is a combination of the previously discussed methods. The following example is worked out in the standard.

Attention! The example is very similar to the estimation method for the purpose of estimating the target SIL, however, it is not completely identical.

Tab3-2/2-Sh.16

Hybrid method

Document no.:
Part of doc. no:

Risk estimation

Product:
Issued by:
Date:

☐ Preliminary risk estimation

Black area = High risk
Grey area = Medium risk
White area = Low risk

Consequences	Severity Se	Class Cl (Fr+Pr+Av) 4	5 - 7	8 - 10	11 - 13	14 - 15	Frequency Fr		Probability Pr		Avoidance Av	
Death, losing an eye or arm	4						≥1 h	5	very high	5		
Permanent, losing fingers	3						< 1 h to ≥ 24 h	5	likely	4		
Reversible, medical attention	2						< 24 h to ≥ 2 w	4	possible	3	impossible	5
Reversible, first aid	1						< 2 w to ≥1 y	3	rarely	2	possible	3
							< 1 y	2	negligible	1	likely	1

Ref. no.	Typ. Hzd. No.	Hazard	Se	Fr	Pr	Av	Cl
1							
2							
3							

Details (description of the accident scenario) of ref. no.

1	
2	
3	

Step 3: Estimating risks

Hybrid method (NPR-ISO/TR 14121-2, paragraph 6.5)

Probability, Pr

Pr is the probability of occurrence of a hazardous event.

Consider, for example, human behaviour, reliability of components, accident history and the nature of the component or system (for example, a knife is always sharp, a flue exhaust pipe is always hot, electricity is dangerous by its nature) to determine the level of probability.

Step 3: Estimating risks

Hybrid method (NPR-ISO/TR 14121-2, paragraph 6.5)

Probability, Pr

The probability is scored as follows:

1. Negligible: for example, this kind of component never fails so that a hazardous event occurs. No possibility of human error.
2. Rarely: for example, it is unlikely this kind of component will fail so that a hazardous event occurs. Human error is unlikely.
3. Possible: for example, this kind of component can fail so that a hazardous event occurs. Human error is possible.
4. Likely: for example, this kind of component will probably fail so that a hazardous event occurs. Human error is likely.
5. Very high: for example, this kind of component is not made for this application. It will fail so that a hazardous event occurs. Human behaviour is such that the likelihood of error is very high

Step 3: Estimating risks

Hybrid method (NPR-ISO/TR 14121-2, paragraph 6.5)

Avoidance, Av

Av is the possibility of avoiding or limiting harm. Consider, for example, whether the machine to be operated by skilled or unskilled persons, how quickly a hazardous situation can lead to harm, and the awareness of risk by means of general information, direct observation or through warning signs, to determine the level of avoidance.

Step 3: Estimating risks

Hybrid method (NPR-ISO/TR 14121-2, paragraph 6.5)

Avoidance, Av

The possibility of avoidance is scored as follows:

1. Likely: for example, it is likely that contact with moving parts behind an interlocked guard will be avoided in most cases should the interlocking fail and the movements continue.
3. Possible: for example, it is possible to avoid an entanglement hazard where the speed is slow and there is sufficient space or otherwise it is easy to avoid moving parts of machinery.
5. Impossible: for example, it is impossible to avoid a sudden appearance of a powerful laser beam, or in case of an explosion.

Tab3-2/2-Sh.21

Step 3: Estimating risks

Hybrid method (NPR-ISO/TR 14121-2, paragraph 6.5)

Class, Cl

Cl is the class. Fr, Pr and Av are the constituent factors that form the probability of occurrence of harm as described in ISO 12100:2010, 5.5.2.3. Each of the three factors should be estimated independently of each other. The worst credible assumption should be used for each factor. Fr, Pr and Av are added together in Cl. The Cl is the sum of Fr, Pr and Av, i.e., Cl = Fr + Pr + Av.

Tab3-2/2-Sh.22

Hybrid method

Risk estimation

Document no.:
Part of doc. no:

Product:
Issued by:
Date:

☐ Preliminary risk estimation

Black area = High risk
Grey area = Medium risk
White area = Low risk

Consequences	Severity Se	Class Cl (Fr+Pr+Av) 4	5 - 7	8 - 10	11 - 13	14 - 15	Frequency Fr		Probability Pr		Avoidance Av	
Death, losing an eye or arm	4						≥1 h	5	very high	5		
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Reversible, medical attention	2						< 24 h to ≥ 2 w	4	possible	3	impossible	5
Reversible, first aid	1						< 2 w to ≥1 y	3	rarely	2	possible	3
							< 1 y	2	negligible	1	likely	1

Ref. no.	Typ. Hzd. No.	Hazard	Se	Fr	Pr	Av	Cl
1							
2							
3							

Details (description of the accident scenario) of ref. no.

1	
2	
3	

Step 3: Estimating risks

Hybrid method (NPR-ISO/TR 14121-2, paragraph 6.5)

Risk estimation

The risk is estimated by using the matrix in the middle of the upper part of the form reproduced on the previous page.
Where the severity, Se, crosses the class, Cl, in the black area, the risk is high.
Where the severity, Se, crosses the class, Cl, in the grey area, the risk is medium.
Where the severity, Se, crosses the class, Cl, in the remaining area, the risk is low.

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